National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Microprocessor Interfacing and Programming	Course Code:	EE3002
Program:	BS Electrical Engineering	Semester:	Fall 2023
Duration:	15 minutes	Total Marks:	10
Quiz Date:	11-Sept 2025	Weight:	2.5
Section:	BEE-5C	Page(s):	1
Exam Type:	Quiz- 1	CLO #	1

Student Name: [redacted] Roll No. [redacted] Section: [redacted]

Instructions:
1. Do not forget to write your Name and Roll Numbers.
2. Solve on the paper and Return.

Question No. 1 (CLO No. 1) **Marks:**

Demonstrate the operation of the PIC microcontroller through a step-by-step analysis of the execution process the given Assembly Language Program (ALP).

WREG: 0, 22H, 88H.

12H: 0, 22H, 44H, 66H

Assembly Language Program

```
line 1 -> MOVLW 0
line 2 -> MOVWF 12H
line 3 -> MOVLW 22H
          ADDWF 12H, F
          ADDWF 12H, F
          ADDWF 12H, F
          ADDWF 12H, W
```

	WREG	12H
1:	0	0
	0	0
	22H	0
	22H	22H
	22H	44H
	22H	66H
	88H.	66H

line 1: move literal 0 into WREG.
line 2: move the literal which is in WREG to File Reg. 12H.
line 3: move literal 22H into WREG.
line 4: Add WREG to File Reg 12H, saving the result in File Reg 12H.
line 5: Add WREG to FileReg 12H, saving the result to FileReg.
line 6: Add WREG to File Reg 12H, saving the result to File Reg.
line 7: Add WREG to FileReg 12H, saving the result to WREG.

10

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Question No. 2 (CLO No. 1) **Marks:**

Determine the memory addresses and codes of the instructions for the ALP mentioned in Q1.

2^n

88 then in binary:

1000 1000

8 bits.

$2^8 = 256$ Bytes.

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